

IncidentIQ

Notebooks Detailed Guide

Multi-Agent AI Platform for DevOps Incident Detection, RCA & Automated Response

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1. Overview of All Notebooks

There are 4 progressively refined notebooks that demonstrate the IncidentIQ system, moving from proof-of-concept to production-aligned code. Each notebook builds upon the previous one, introducing new concepts and refining implementations.

Notebook	File Size	Purpose	Status	Focus
v1: Original MVP	288 KB	Foundation & Concepts	First Generation	Agent fundamentals
v2: LangGraph	50 KB	Graph Orchestration	Second Generation	LLM + state mgmt
v3: src/ Aligned	99 KB	Architecture Validation	Third Generation	Modular structure
v4: Production	111 KB	Final Polished	Latest (RECOMMENDED)	Best practices

2. Multi_Agent_DevOps_Incident_Analysis_Suite_Colab.ipynb

(288 KB) — Original MVP / Proof-of-Concept

Purpose: First generation implementation; foundation for later versions

Status: MVP, notebook-first approach

Key Demonstrations:

- End-to-end incident analysis pipeline using 9 specialized agents
- Raw Colab implementation without LangGraph orchestration
- Sequential agent workflow (Parser → Signal → Classify → RCA → Timeline → Reporting)
- Simple deterministic scoring without LLM reasoning
- Useful for prototyping and learning core concepts

The 9 Agents:

Agent	Responsibility
Parser	Normalize raw log lines into structured events
Signal Extraction	Detect operational signals using regex patterns
Classifier	Infer incident category (Database, Network, Auth, etc.)
RCA	Perform root cause analysis with supporting evidence
Timeline	Reconstruct incident chronology from events
Notification	Generate Slack/Teams-style alert previews
Ticket	Generate JIRA/GitHub-style ticket previews
Cookbook	Create reusable runbook/checklist items
Eval/Scoring	Compare results against hidden ground truth

Workflow:

1. User uploads raw log files (.log, .txt, .jsonl, .zip)
2. Parser normalizes them into structured LogEvent objects
3. Signal extractor finds operational anomalies using pattern matching
4. Classifier + RCA agents reason over evidence
5. Timeline agent builds incident narrative
6. Agents generate preview notifications and tickets
7. Eval module compares results to ground truth

3. Multi_Agent_DevOps_Incident_Analysis_Suite_LangGraph_Colab_v2.ipynb

(50 KB) — LangGraph + LLM Integration

Purpose: Second generation; introduces graph orchestration and LLM reasoning

Status: Adds structured state management and API integration

What It Adds:

- **LangGraph Integration** — Replaces simple sequential calls with a proper workflow graph
- **LLM Reasoning Layer** — Moves ambiguous logic to Claude/GPT instead of heuristics only
- **Dual-Mode Operation** — No-key mode (deterministic) + LLM mode (enhanced reasoning)
- **State Management** — IncidentGraphState object passed between graph nodes
- **Conditional Routing** — Retry loops, fallback paths, critic feedback

Key Sections:

Section	Focus
LLM Configuration	API key setup, model selection, fallback mode
Upload Logs	Colab file picker integration
Schemas	Pydantic models for state passing
Signal Extraction & Clustering	Rules-based anomaly detection + grouping
RAG Library	Embedded runbook retrieval (lexical search)
LangGraph Agents	Agents that reason with LLMs
Run Pipeline	Execute the LangGraph workflow
Eval & Scoring	Compare results to ground truth

Key Improvements:

- Introduces IncidentGraphState (shared state object)
- Agents are now LangGraph nodes that transform state
- Explicit state passing between agents
- LLM-driven reasoning for ambiguous cases
- Critic agent validates conclusions to reduce hallucinations

4. Multi_Agent_DevOps_Incident_Analysis_Suite_SRC_Aligned_Colab_v3.ipynb

(99 KB) — src/ Architecture Aligned

Purpose: Third generation; aligns notebook structure with production code organization

Key Innovation: Single comprehensive notebook where each section maps directly to a src/ folder module

Section-by-Section Mapping to src/:

Notebook Section	src/ Folder	Purpose
config/	src/config/	Settings, feature flags, LLM API config
models/	src/models/	Pydantic enums, LogEvent, Cluster, Incident
utils/	src/utils/	Hash, timestamps, paths, text utilities
ingest/	src/ingest/	ZIP loader, log parser, normalizer
signals/	src/signals/	SignalRule engine, pattern matching
clustering/	src/clustering/	EvidenceClusterer, hybrid strategies
rag/	src/rag/	RunbookStore, lexical retriever
agents/	src/agents/	LLM agents (Classifier, RCA, Timeline, etc.)
graph/	src/graph/	LangGraph state, nodes, builder, runner
reporting/	src/reporting/	Markdown, JSON, CSV, previews, bundles
evals/	src/evals/	GroundTruthLoader, scorer, metrics

Additional Sections:

- **End-to-end runner** — Full pipeline orchestration
- **Colab batch execution** — Run multiple datasets
- **Smoke tests** — Validate on sample data
- **Inspect results** — Debug incidents and scores
- **Download bundle** — Export all outputs
- **Refactor checklist** — Notes on moving code to src/

Why This Matters:

v3 is intentionally modular — each cell's code is organized exactly as it should be in src/. This makes it straightforward to refactor: copy working code from a notebook cell → move to corresponding src/ file → test →

integrate. The notebook serves as a bridge between exploratory prototyping and production code.

5. Multi_Agent_DevOps_Incident_Analysis_Suite_SRC_Aligned_Colab_v4.ipynb

(111 KB) — Production-Ready Polished Implementation

Purpose: Latest, most complete reference implementation

Status: RECOMMENDED for use and as a reference

What It Improves: Bug fixes, enhanced diagnostics, better error handling

Improvements from v3 to v4:

- **v4 Stabilization Focus** — Production-grade refinements and bug fixes
- **Better Diagnostics** — More detailed mismatch analysis in eval section
- **LLM Connectivity Test** — Optional cell to validate OpenAI/OpenRouter setup
- **Enhanced Inspection** — Eval mismatch diagnostics understand why incidents don't match
- **Cleaner Code** — Refactored for clarity and maintainability
- **Robust Error Handling** — Better exception handling in ingest and clustering
- **Improved Evidence Citing** — RCA output includes better source references

New/Enhanced Features:

Cell	Enhancement
14b	Optional LLM connectivity smoke test
16	Mismatch diagnostics: which incidents detected vs. ground truth?
	Which categories were misclassified? Which severities wrong?
	Which clusters should have been merged/split?

Recommended Usage:

- **For learning:** Use v4 as the reference implementation
- **For refactoring:** Copy code from v4 sections directly into src/ modules
- **For testing:** Run v4 with sample datasets to validate behavior
- **For extending:** Add new agents or reporters to v4 first, then migrate to src/

6. Detailed Feature Breakdown

Core Components (All 4 Notebooks Have These):

Input Validation & File Loading

.zip extraction (safe), .jsonl/.log/.txt parsing, skip eval-only folders at runtime

Log Normalization

Parse raw lines into LogEvent objects with fields: timestamp, service, level, trace_id, message

Signal Extraction

Regex-based rules for database errors, network timeouts, auth failures; signal weights 0.0-1.0

Clustering/Deduplication

Group related signals into incident candidates; prevent duplicate reports

Agent-Based Reasoning

Classifier (category & severity), Timeline (chronology), RCA (root cause), Remediation, Critic

Output Generation

Markdown reports, JSON/CSV exports, Slack/Teams previews, JIRA/GitHub previews, cookbooks

Evaluation/Scoring

Load ground truth, score category accuracy, severity accuracy, evidence grounding

7. Data Flow Across Notebooks

Step 1: Input

User uploads logs (ZIP or plaintext files)

Step 2: Ingest

Parse & normalize into LogEvent objects

Step 3: Signal Detection

Deterministic pattern matching extracts signals

Step 4: Clustering

Group evidence into incident candidates

Step 5: Agent Reasoning

LLM agents (with fallback to rules) perform:

- Classification (category, severity)
- Timeline building
- Root cause analysis
- Remediation recommendations
- Critic review (validate & reduce hallucinations)

Step 6: Reporting

Generate Markdown reports, JSON, CSV, preview notifications/tickets

Step 7: Evaluation

Score against ground truth (eval-only step, isolated from runtime)

Step 8: Download

User downloads bundle with all outputs

8. How to Use These Notebooks

If You Want To...	Use This Notebook	Why
Learn the core concepts	v1 (Original MVP)	Simplest agent structure, no graph complexity
Understand LangGraph + LLM	Mv2 (LangGraph)	Introduces orchestration and state management
Refactor code to src/	v3 (src/ Aligned)	Clear 1:1 mapping between cells and src/ files
Run in practice	v4 (Production-Ready)	Latest, most stable, best diagnostics

Workflow for Development:

1. **Understand:** Read the notebook cell-by-cell to understand the logic
2. **Modify:** Change code and test in Colab to validate behavior
3. **Extract:** Move working code from notebook cell to corresponding src/ file
4. **Integrate:** Import from src/ back into notebook for validation
5. **Test:** Run full pipeline to ensure integration works correctly

9. Key Takeaway

v4 is the comprehensive reference implementation.

Each section of v4 directly maps to a `src/` module, making it straightforward to:

1. **Understand** — Read the notebook cell-by-cell
2. **Modify** — Change logic and test in Colab
3. **Extract** — Move working code to `src/` files
4. **Integrate** — Import from `src/` back for validation

The notebooks are intentionally modular. They serve as a bridge between:

- Exploratory prototyping (Google Colab environment)
- Production code (clean `src/` modules)

This design allows for rapid iteration and experimentation while ensuring that eventually, the code is refactored into a maintainable, testable, production-grade codebase.

Happy Hacking! ■